

SEFI:DNA

Identity Geometry for Multi-Entity Systems

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Abstract

SEFI:DNA extends the Single Entity Field Interpretation (SEFI) by introducing a geometric framework for multi-entity identity. While SEFI Space defines a four-dimensional identity manifold for a single entity, SEFI:DNA constructs a tensor product identity space capable of representing coupled identity trajectories, identity entanglement, and multi-entity stability. The resulting structure provides a geometric interpretation of multi-particle identity, entanglement, and transformation compatible with worldline quantum field theory and SEFI/GWFM geometry.

1 Introduction

SEFI Space defines identity geometry for a single underlying entity using four basis directions: FIELD ORIGIN, FIELD AUTHORSHIP, FIELD SOVEREIGNTY, and WARP EXPRESSION. The identity state evolves as a trajectory in SEFI Space while the worldline traverses spacetime.

Multi-entity systems require a generalization of this identity geometry. SEFI:DNA introduces a multi-dimensional identity space constructed from tensor products of SEFI Space, allowing identity coupling, identity entanglement, and multi-entity sovereignty constraints.

The goals of this manuscript are:

- To define SEFI:DNA as a tensor product identity space for multi-entity systems.
- To introduce identity entanglement and coupled identity trajectories.
- To define multi-entity sovereignty and warp constraints.
- To couple SEFI:DNA to worldline geometry in a manner compatible with SEFI/GWFM.

2 Multi-Entity Identity Space

Let SS denote SEFI Space for a single entity. For a system of two entities, we define the tensor product identity space

$$SS_2 = SS \otimes SS.$$

An identity state in SS_2 is represented by

$$\mathbf{I}_2(t) = \mathbf{I}_1(t) \otimes \mathbf{I}_2(t),$$

where each $\mathbf{I}_i(t)$ is a SEFI Space identity vector.

For N entities, the identity space becomes

$$SS_N = SS^{\otimes N}.$$

Identity coordinates for each entity combine to form a multi-entity identity tensor.

3 Identity Entanglement

Identity entanglement occurs when the identity state cannot be written as a product of individual identity vectors:

$$\mathbf{I}_2(t) \neq \mathbf{I}_1(t) \otimes \mathbf{I}_2(t).$$

Entanglement corresponds to coupled identity trajectories in SEFI:DNA. These coupled trajectories represent geometric identity correlation between entities.

Identity entanglement may arise from:

- geometric overlap of worldlines,
- shared sovereignty constraints,
- coupled warp transformations,
- multi-entity stability surfaces.

4 Multi-Entity Sovereignty

Sovereignty generalizes to multi-entity systems by requiring invariance of the identity tensor under allowed multi-entity transformations T :

$$\mathbf{I}_N(t) = T(\mathbf{I}_N(t)).$$

The allowed transformations form a sovereignty-preserving subgroup of the full transformation group acting on the multi-entity worldline geometry.

Multi-entity sovereignty defines stability surfaces in SS_N , and errors correspond to displacements orthogonal to these surfaces.

5 Multi-Entity Warp

Warp operations generalize to multi-entity systems by acting on each identity coordinate and on coupled identity components.

A multi-entity warp operator W_N acts on the identity tensor as

$$\mathbf{I}'_N(t) = W_N(\mathbf{I}_N(t)).$$

Coupled warp operations may induce correlated identity transformations, producing motion in SS_N along multi-axis directions.

Warp must satisfy multi-entity sovereignty constraints to preserve identity.

6 Coupling to Worldline Geometry

Identity coordinates in SEFI:DNA are constructed from multi-entity worldline geometry. For two entities with worldlines $r_1(t)$ and $r_2(t)$, identity coordinates may be defined as

$$I_O^{(1)}(t) = O(r_1(t)), \quad I_O^{(2)}(t) = O(r_2(t)),$$

and similarly for authorship, sovereignty, warp, and reality-layer functions.

Coupled worldline geometry induces identity entanglement and multi-entity warp interactions.

7 Discussion and Outlook

SEFI:DNA provides a geometric identity framework for multi-entity systems. By defining identity tensors, identity entanglement, multi-entity sovereignty, and multi-entity warp, the framework extends SEFI Space to systems with multiple interacting entities.

Future work will focus on:

- deriving explicit multi-entity layer functions,
- exploring identity geodesics in SS_N ,
- defining multi-entity stability surfaces,
- investigating applications to quantum error correction and entanglement geometry.